

1 Quick Review: Free-Body Diagrams

1.1 What is a free-body diagram?

A free-body diagram represents the object as a particle and shows all of the forces acting on the object. According to your textbook it is developed by:

1. Identify **ALL** forces acting on the object.
2. Draw a coordinate system.
3. Represent the object as a dot at the origin of the coordinate axes.
4. Draw vectors representing **EACH** of the identified forces.
5. Draw and label the net force vector $\vec{F}_{net}$ next to the diagram.

I will omit any draw diagrams in this section as they will be thoroughlt covered in the practice problems.

1.2 What are the different types of forces?

The textbook identifies EIGHT different forces:

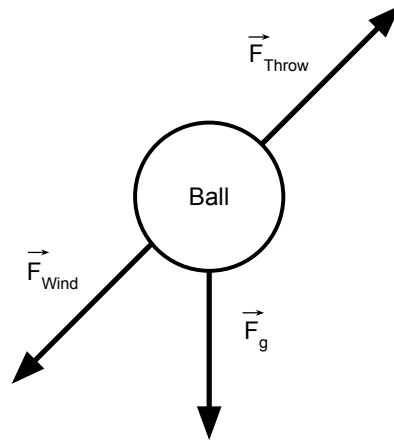
1. **Gravity**
2. *Spring Force*
3. *Tension Force*
4. **Normal Force**
5. **Friction**
6. **Drag**
7. *Thrust*
8. *Electric & Magenetic Forces*

I have went ahead and boldened the forces we will be working with today. The only forces you will not cover in 220 are the Electric & Magenetic Forces.

2 Practice Problems

2.1 Identifying Forces & Free-Body Diagrams

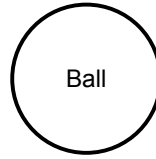
1. **(Visualize)** A man throws a ball in the air. Unfortunately, the wind is not in his favor. Identify what would happen to the ball in this scenario based on its free body diagram.



2. **(Visualize)** Indicate the angles the force from the wind and the force from the throw are from the horizontal using θ to represent the angle.
3. **(Apply)** Given the free body diagram provided in (part 1) and built upon in (part 2), create an equation representing the forces acting on the ball, F_{net} .

$$\sum F_{\text{net}} =$$

4. **(Visualize)** Immediately after his friend, who is facing the other way, throws a ball of his own. Fortunately, the wind is in his favor. Create a free body diagram for this scenario. Use θ to represent the angles between $\vec{F}_{\text{wind}}$ and $\vec{F}_{\text{throw}}$ and the horizontal, respectively.



5. **(Explain)** Why will his friend appear to have a much stronger throw in this scenario?

6. **(BONUS)**

- (a) Write equations for (part 4) that would represent, F_x and F_y , the forces in the x - and y -axis directions, respectively, acting on the ball.
- (b) Use the equations for F_x and F_y you found in (part a), $\theta = 30^\circ$, and $\|\vec{F}_{wind}\| = \|\vec{F}_{throw}\| = 40 \text{ N}$ to find the magnitude of the force.

2.2 Dynamics

1. **(Conceptualize)** Discuss the difference among:

- (a) Mass & Weight

- (b) Equilibrium & Constant-Force

- (c) Static Friction & the Coefficient of Static Friction

- (d) Acceleration & Force

2. **(Derive & Apply)** The force on a body is represented by Newton's Second Law as $\vec{F} = m\vec{a}$. A box that weighs 32 N is slid across a horizontal surface by a rope, $\vec{F}_{rope}$. This force consists of a constant acceleration of 3.0 m/s^2 .

- (a) Identify all the forces acting on the box when negating friction and draw a free-body diagram.

- (b) Find the mass of the box.

- (c) Now consider that a friction $\vec{f}$ was present. Draw a free body diagram for the new situation. Find a way to solve for the coefficient of friction μ .

- (d) Now consider that when $\vec{F}_{rope}$ is opposed by friction f , the acceleration drops to 2.0 m/s^2 . Use your result from part (c) to find the coefficient of friction μ required for this to happen.

- (e) **(BONUS)** Acceleration can be represented by $a(t) = \frac{dv}{dt}$. Given the acceleration from (part d), find the velocity as a function of time, $v(t)$.

3. **(Conceptualize, Derive, & Apply)** Drag is the force that opposes the motion of an object moving through a fluid (such as air or water).

- (a) What is the difference between a high Reynold's number and a low Reynold's number?

- (b) Using $\|F_{drag}\| = \frac{1}{2}C_d\rho Av^2$ find the terminal speed v_{term} of a sphere with radius r and mass m . $C_{d,sphere} = 0.50$, $\rho_{air} = 1.2 \text{ kg/m}^3$.

- (c) Using your result from (part b) find v_{term} of a sphere with radius $r = 300 \text{ cm}$ and mass:

$$m = 5 \text{ kg}$$

$$m = 10 \text{ kg}$$



- (d) Does the density of an object seem to impact the terminal speed of a sphere? If it does, explain how by determining if the speed seems to decrease/increase quickly or slowly as the density increases?
